

Metadata-Guided MoErging of Generative Models for Privacy-Preserving Domain Adaptation in Medical Image Analysis

Project vision: The vision of this project is to reshape healthcare data-sharing by replacing the exchange of sensitive, high-volume datasets with privacy-preserving, efficient, and personalized generative models. Rather than sharing large-scale raw clinical images and associated personal health information (PHI), institutions will exchange lightweight models capturing local data distributions while offering formal privacy guarantees. As illustrated in Figure 1, the proposed approach allows sampling (i.e., generating) synthetic data tailored to new and previously unseen domains while rigorously protecting patient privacy. Furthermore, its modular designs support the addition or removal of institutions with minimal overhead, as no end-to-end training across sites is required.

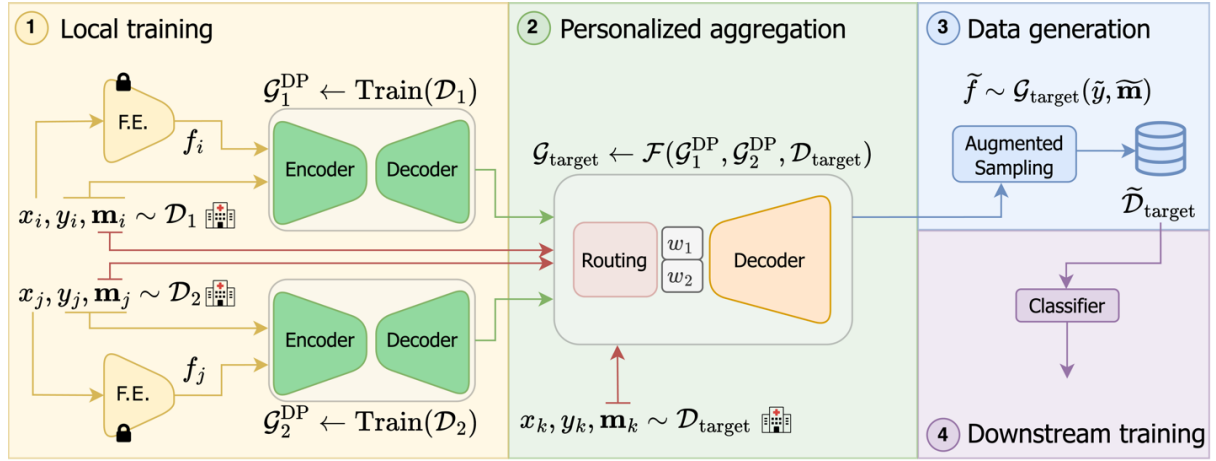


Figure 1: Overview of SynGenacy, a modular generative framework enabling privacy-preserving, domain-adapted, and efficient synthetic data generation across multiple institutions. (1) Each institution $h \in \{1, \dots, H\}$ independently trains a differentially private (DP) generative model $\mathcal{G}_h^{\text{DP}}$, thereby providing formal privacy guarantees, on its private dataset $\mathcal{D}_h = \{(x_i, y_i, \mathbf{m}_i)\}_{i=1}^{N_h}$, where N_h defines the number of samples and $\mathbf{m}_i$ encodes structured metadata (e.g., scanner type, resolution, patient information) associated with image x_i and label y_i . The raw image x_i is first processed through a large pre-trained feature extractor (F.E.) to obtain a compact and informative latent representation f_i . (2) A target institution provides a support set $\mathcal{D}_{\text{target}}$ from which metadata statistics are used to compute similarity scores with respect to the source domains. These scores define soft routing weights over the pretrained generators, enabling the construction of a personalized generator $\mathcal{G}_{\text{target}}$. Notably, this aggregation is performed post-hoc. Therefore, during training, each model remains fully local. (3) The personalized generator $\mathcal{G}_{\text{target}}$ is used to synthesize a diverse synthetic target-specific dataset $\tilde{\mathcal{D}}_{\text{target}}$ via an augmented sampling. (4) Lastly, the synthetic personalized dataset supports downstream classifier training without access to raw images or personal health information (PHI), ensuring data privacy.

Scope of the thesis: In multi-institutional medical imaging, models trained independently on local datasets often capture complementary but domain-specific variations (eg, scanner type, staining protocols, patient population, and more). Integrating these models into a unified system without data sharing is crucial for scalability and privacy preservation.

This thesis explores a novel metadata-driven model merging approach inspired by recent work on model MoErging [1], a paradigm combining ideas from model merging [2] and Mixture-of-Experts (MoE) [3,4]. Unlike traditional model fusion (eg, weight interpolation or knowledge distillation), MoErging enables post-hoc, decentralized, and privacy-aware model integration by leveraging structured metadata to guide model selection at inference time. To facilitate comparison and clarify assumptions, we adopt a recent taxonomy proposed in [1], which categorizes MoErging design choices along three core dimensions: (i) the experts , which are independently trained generative models shared by contributors in a decentralized setting; (ii) the routing strategy , which determines how experts are selected and potentially aggregated based on task or domain metadata; and (iii) the downstream application , which specifies how the composed model is used, eg, image generation, augmentation, or diagnosis support. This framework enables the systematic exploration of personalized model composition without data centralization or expert retraining.

Therefore, given a set of pre-training datasets (from different institutions) and a target dataset, the aim is to independently train generative experts on each source domain and later compose a personalized generator by selecting and aggregating the most relevant experts based on their affinity to the target domain, measured through structured metadata or via dataset's numerical statistics.

Research Objectives

1. *Define a rigorous evaluation framework based on available datasets*

- Explore the available datasets listed in the next section and define a “priority” list, indicating the most suitable ones for the current evaluation.
 - *The list is surely not exhaustive and novel datasets come out quite frequently. In case we find further nice datasets, we can consider them.*
- Based on the shortlisted datasets, define a rigorous evaluation framework, including number of experts (based on metadata), respective distribution shifts, downstream evaluation metrics, and so on.

2. *Develop conditioned Generative Models*

- Develop conditioned generative models for privacy-preserving data sampling by adopting foundation models for feature extraction as in [5,6].
- These models will be conditioned on demographic and acquisition-related attributes to enable controlled and fair sampling across subgroups.

3. *Design a Metadata-Based Routing Mechanism*

- Define domain similarity metrics using structured metadata (e.g., patient cohort statistics, scanner vendor, resolution, staining protocol) or using raw-data statistics.
- Develop a routing mechanism that dynamically selects a relevant subset of experts for each generation task.

4. *Construct the MoErging Generator*

- Use independently trained generative models (e.g., Conditional VAEs [5,6]) for each source domain.
- Develop an inference-time composition method that routes generation requests through selected experts without retraining or parameter sharing. A possible baseline is given by FedAvg on experts’ decoders [6].

5. *Evaluate domain-specific and generalization performance*

- Assess generation quality and downstream classification performance.
- Benchmark across unseen test domains with varying metadata overlap.
- Compare against baselines such as naive ensembling and weight averaging

Positioned at the intersection of generative models, model aggregation, mixture-of-experts , and privacy , this project contributes to multiple active research areas and holds strong potential for a publishable outcome!

Datasets

- [MIDOG++](#) [7], distributed under MIT License. This dataset provides explicit metadata on scanner, laboratory, and staining protocols. A clear split separation is further defined in [OpenMIBOOD](#) [8].
- [Camelyon17](#) [9,10] shared under Creative Commons Zero (CC0) License. This is an established dataset including slides acquired from five different hospitals.
- [BCNB](#) [11], custom license, available for academic research use. It comprises patient- and study-level clinical metadata.
- [BreakHis](#) [12], distributed under the Creative Commons Attribution 4.0 International License (CC BY-NC 4.0). It includes slides acquired with different magnification factors (40×, 100×, 200×, 400×).
- **(optional)** [PathMNIST-C](#) [13], distributed under the Creative Commons (CC) License. It includes real-world corruption types utilized for benchmarking downstream model robustness against distribution shifts.
 - **Optional** - Another rigorous evaluation and stress test protocol for our routing can be via controlled data *with synthetic shifts*, where distribution shifts are induced by image-based realistic corruptions (Di Salvo, Doerrich, & Ledig, 2024) and the corresponding augmentation parameters are encoded as metadata. This will enable controlled robustness stress tests and a comprehensive validation of our routing and aggregation strategies in a setting where the metadata-shift relationship is fully known.

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